Bayesian inference of frictional parameters from seismograms*

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Abstract. *At most 150 words: the application and the inverse problem, the approach (model, prior, inference method), the main result in one sentence, and one sentence on uncertainty or limitations. Write it last.*

We consider ...

Key words. inverse problems, Bayesian inference, uncertainty quantification, *three to five keywords*

Code repository. https://github.com/robindymer/bayesian-inverse-problems

1. Introduction.

About 1–1.5 pages. Answer four questions. (i) What is the application, and why is the task an inverse problem? (ii) How does it connect to your own research? (iii) What has been done before? Cite 5–10 references, e.g., textbooks such as [3, 1] and the key papers of your field. (iv) What exactly did you do? State your contributions in two or three sentences or as a short list. A one-sentence outline of the report is optional.

Inverse problems arise whenever an unknown quantity has to be recovered from indirect and noisy observations [3, 5]. In this report, we ...

The contributions of this report are as follows.

- We formulate ... as a Bayesian inverse problem with ...
- We solve it using ... and verify the implementation on synthetic data.
- We quantify the uncertainty in ... and compare with ...

2. Problem formulation.

About 2.5–4 pages in total for this section. It must contain the forward model and the data, a demonstration of ill-posedness, the likelihood, the prior, the posterior (or the regularized formulation and its Bayesian interpretation), and an explicit list of assumptions. This is the section the report discussion in December will start from.

2.1. Forward model and data.

What is unknown, what is observed, and how are the two related? Give the forward model (continuous and discretized), the dimensions, and the data: what is measured, how many measurements, how noisy, real or synthetic. Show that the problem is ill-posed, for instance by a naive inversion that fails, by the decay of singular values, or by a non-identifiability argument.

2.1.1. 1.5.

The unknowns of our problems are the frictional parameters that exist in the fault. Two of the more important once is the direct-effect a and the state evolution parameter b, where their difference $a-b$ determines the possibility of unstable rupture. If $a-b > 0$, friction increases with slip velocity, known as velocity-strengthening. If instead $a - b < 0$, friction decreases with slip velocity, which is unstable and known as velocity-weakening. Following

*Final-project report for the PhD course *Computational and Bayesian Inverse Problems*, Linköping University, autumn 2026. Submitted October 29, 2026.

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the work in [4], we focus on the case of inverting for the single parameter $a(x)$ which allows us to dodge the problem of needing regularization schemes that handle parameter trade-offs. But for generality, we denote the unkown that we unknown that we are inverting for as p, which can be thought of being e.g. a or b. The forward model is given by

$$
\begin{aligned}
\rho \ddot{u}_J &= \partial_I C_{IJKL} \partial_K u_L + Q_J, && \overline{x} \in \Omega, && t \in \mathcal{T}, \\
\overline{u} &= \overline{u}_0, \quad \dot{\overline{u}} = \overline{v}_0, && \overline{x} \in \Omega, && t = 0, \\
L\overline{u} &= \overline{g}, && \overline{x} \in \partial\Omega, && t \in \mathcal{T}, \\
V_n &= 0, && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\overline{T}^+ &= -\overline{T}^-, && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\tau_J &= F_J(\overline{V}, \Psi, p), && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\dot{\Psi} &= G(\overline{V}, \Psi, p), && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\Psi &= \Psi_0, && \overline{x} \in \Gamma, && t = 0.
\end{aligned}
\tag{2.1}
$$

Explain variables etc. The data is usually from seismograms, where we of course have sources of noise (measurement error, scattering). There is also some modelling error which also can be though of as noise. It is standard to model this noise as additive gaussian, which we therefore do in this report. Let $\boldsymbol{p} \in \mathbb{R}^n$ denote the unknown and $\boldsymbol{y} \in \mathbb{R}^m$ the observations. We assume the observation model

$$
\boldsymbol{y} = \Phi(\boldsymbol{p}) + \boldsymbol{e}, \tag{2.2}
$$

where Φ is the forward operator that describes how the unknown produces noise-free data and $\boldsymbol{e}$ is measurement noise. Usually we measure particle velocity, i.e. $\boldsymbol{y} = \dot{\boldsymbol{u}}$, sampled at the recievers. The forward map is not linear due to the non-linear friction law at the fault. The forward problem is given $\boldsymbol{p}$, compute $\Phi(\boldsymbol{p})$ which we cannot express as $\Phi(\boldsymbol{p}) = A\boldsymbol{p}$. Coefficient problems are nonlinear even when the physics is linear, because the state depends nonlinearly on the coefficients. E.g. $\boldsymbol{a}$ multiplies with V and Ψ, that themselves depend on $\boldsymbol{p}$. Of Hadamard's conditions, we expect all three to cause difficulties. *Existence*: well-posedness of the forward problem is only established in special cases, such as 2D antiplane shear with admissible state evolution laws [4]. For the inverse problem, noisy data $\boldsymbol{y}$ almost surely lie outside the range of Φ, so no $\boldsymbol{p}$ reproduces them exactly; we therefore minimize the misfit $\frac{1}{2}\|\Phi(\boldsymbol{p}) - \boldsymbol{y}\|^2$ instead. *Uniqueness*: in velocity-strengthening regions the fault does not slip and hence does not radiate, so the data are insensitive to p there and different values give the same seismograms. Moreover, the misfit is non-convex, and inversions can stagnate in local minima. Inverting for several parameters at once would add trade-offs, e.g. between $a - b$ and the initial shear stress. *Stability*: the data are band-limited and recorded at a finite number of receivers away from the fault, so fine-scale variations in p barely affect them. Refining the parameter grid indeed produces spurious oscillations in the reconstructed p, so the coarse parametrization acts as a regularization, and further regularization is likely needed for real data. One evaluation of the forward map is one time integration of the SBP-SAT finite difference scheme with RK4, including a bisection solve for the slip velocity V^* at each fault point and Runge–Kutta stage; its cost grows with the number of grid points and time

steps. The full Jacobian $J \in \mathbb{R}^{m\times n}$ requires one linearized forward solve per unknown p_i or one adjoint solve per data sample y_j, i.e. $\min(n, m)$ solves. Since $m = N_{rec}N_t \gg n$, the n linearized forward solves are far cheaper. This is affordable in the setting of [4], where the fault is one-dimensional and p is represented on a coarse grid with $n = m_p$ points (e.g. $m_p = 26$), but becomes prohibitive for 3D faults with many unknowns. Gradient-based optimization, however, only needs the misfit gradient $J^\top \boldsymbol{r}$, with residual $\boldsymbol{r} = \Phi(\boldsymbol{p}) - \boldsymbol{y}$, which the adjoint method provides at the cost of a single additional solve, independent of n.

2.1.2. 2.6. The regularized functional becomes

$$J_\lambda(\boldsymbol{p}) = \frac{1}{2}\|\Phi(\boldsymbol{p}) - \boldsymbol{y}\|_2^2 + \lambda R(\boldsymbol{p}), \tag{2.3}$$

where the regularized solution is

$$\boldsymbol{p}_\lambda \in \arg\min_{\boldsymbol{p}\in\mathbb{R}^n} J_\lambda(\boldsymbol{p}). \tag{2.4}$$

For our earthquake model, the frictional parameters are often assumed to be piecewise constant source?. Therefore, we use total variation (TV) regularization, $R(\boldsymbol{p}) = \|L\boldsymbol{p}\|_1$, where L computes the variation of $\boldsymbol{p}$. This will promote sharp corners and non-oscillating solutions. Since the forward map is not linear, we compute the gradient via the adjoint-state method. Since Gauss–Newton requires the full Jacobian, the cost for one iteration is $\min(n, m)$ linearized forward solves with one added forward solve for the residual calculation. Since we use synthetic data, the nature of the noise is known and we therefore know σ. Since the main weakness of this method is an inaccurate estimate of σ, it fits well for this problem. The L-curve method would be very expensive since many inversions would need to be done, one for each λ value. Generalized cross-validation is also expensive.

2.1.3. 3.6. We assume Gaussian noise for our synthetic data. The covariance / noise level Γ is $\sigma^2 I$, which is true for independent receivers (in our case, independent both in space and time). In the case of synthetic data, Γ come from how we generate the data. Then we simply set σ and therefore also Γ. In contrast, for real world data, this would come from the specifications of the measuring equipment. The prior that best fits our belief of the unknown $\boldsymbol{p}$ is the hierarchical prior since it promotes piecewise constant signals. This type of prior is however a bit more expensive than e.g. a smooth Gaussian prior since it uses a hyper-prior for its covariance, and a sample is drawn in two steps. TODO: Plot and determine if samples look plausible. But because of the added complexity of the hierarchical prior, we start with a smoothness prior for the implementation. As for the hyper-parameters of our model, we have several in the Bayesian setting. These are noise level σ, the prior scale γ, and ϑ (and β) for the hierarchical prior. σ comes from how we generate the data, and γ or ϑ is from what we know about $\boldsymbol{p}$. Going with the values in [4], we have that $a(x)$ is between 0.009 and 0.013, therefore the jump size is 0.004, and there are two of those jumps. Note that ϑ sets that amplitude of the jumps, β controls the tail, γ is the smoothness prior and the prior mean we set reasonably at 0.011.
Since we want to start simple, and then move on to the most promising prior (hierarchical). We have that

$$\mathbf{a} \sim \mathcal{N}(\mu, C), \tag{2.5}$$

where $\mu = (0.013 + 0.009)/2 = 0.011$ and $\gamma = 0.002$, i.e. the standard deviation is of the distance between the mean value and min / max value.

2.2. 4.6. What we have is a nonlinear forward operator, and the gradient of the misfit via the adjoint method. We could therefore use MALA or HMC. This will be compared against pCN, which is prior-preserving and does not utilize the gradient. So we are in category three of page 56. One forward evaluation can cost seconds or minutes, depending on the mesh size used. One gradient evaluation costs one forward and one adjoint solve, where the adjoint solve is approximately as expensive as one forward solve. What could be interesting is the pointwise posterior mean and standard deviation of $\boldsymbol{a}$, and the probablity that $a < b$ at the node. A few hundred samples is probably reasonable for us. The cost of Table 4.2 sets the limit. The more samples we can use, the more confident we can be in our estimations. The cost per sample according to our table and using $m_p = 26$ is in our case 7000 for HMC. pCN with one forward solve per itertion would cost about 30 000. The sampler can be verified by simply plotting the sampled posterior and see if it matches the true value well enough.

2.3. Bayesian formulation. *Noise model → likelihood. Prior: which knowledge does it encode (smoothness, sparsity, positivity, physical bounds), and what are its hyperparameters? Posterior by Bayes' rule. If you work with a regularized formulation instead, state the functional and its interpretation as a MAP estimate. If you use a hierarchical model, say which hyperparameters are inferred and which are fixed.*

Modelling the noise as $\boldsymbol{e} \sim \mathcal{N}(\boldsymbol{0}, \sigma^2 \boldsymbol{I})$ gives the likelihood

$$\pi(\boldsymbol{y} \mid \boldsymbol{u}) \propto \exp\Big(-\tfrac{1}{2\sigma^2} \|\boldsymbol{A}\boldsymbol{u} - \boldsymbol{y}\|_2^2\Big). \tag{2.6}$$

We encode prior knowledge about ... through the Gaussian prior

$$\pi(\boldsymbol{u}) \propto \exp\Big(-\tfrac{\delta}{2} \|\boldsymbol{L}\boldsymbol{u}\|_2^2\Big), \tag{2.7}$$

where $\boldsymbol{L}$ is ... and $\delta > 0$ controls ... By Bayes' rule, the posterior is

$$\pi(\boldsymbol{u} \mid \boldsymbol{y}) \propto \pi(\boldsymbol{y} \mid \boldsymbol{u})\, \pi(\boldsymbol{u}). \tag{2.8}$$

The maximum a posteriori (MAP) estimate

$$\boldsymbol{u}_{\mathrm{MAP}} = \arg\min_{\boldsymbol{u}} \ \tfrac{1}{2\sigma^2} \|\boldsymbol{A}\boldsymbol{u} - \boldsymbol{y}\|_2^2 + \tfrac{\delta}{2} \|\boldsymbol{L}\boldsymbol{u}\|_2^2 \tag{2.9}$$

coincides with Tikhonov regularization [2], which we use as a baseline in section 4.

2.4. Assumptions. *List every modelling assumption and say in one clause why it is reasonable or where it is questionable. Typical items: additive Gaussian noise with known or unknown variance, independence of measurements, correctness of the forward model (no model error), the prior family and its hyperparameters, boundary conditions, discretization fine enough to neglect discretization error.*

Assumption 2.1. The noise is additive, Gaussian, and independent across measurements, with known variance σ^2. This is justified because ...

Assumption 2.2. The forward model (2.2) is exact up to discretization; model error is neglected. This is questionable when ...

Algorithm 3.1 Random-walk Metropolis for the posterior (2.8) (replace with your method)

Require: initial state $\boldsymbol{u}^{(0)}$, proposal scale $\tau > 0$, number of samples N

1: **for** $k = 1, \dots, N$ **do**
2: propose $\boldsymbol{u}^\star = \boldsymbol{u}^{(k-1)} + \tau \boldsymbol{\xi}$ with $\boldsymbol{\xi} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$
3: compute $\alpha = \min\{1,\ \pi(\boldsymbol{u}^\star \mid \boldsymbol{y}) / \pi(\boldsymbol{u}^{(k-1)} \mid \boldsymbol{y})\}$
4: set $\boldsymbol{u}^{(k)} = \boldsymbol{u}^\star$ with probability α, otherwise $\boldsymbol{u}^{(k)} = \boldsymbol{u}^{(k-1)}$

Ensure: samples $\boldsymbol{u}^{(1)}, \dots, \boldsymbol{u}^{(N)}$

Table 1: Example table: comparison of methods on synthetic data with 5 % noise. Replace with your own results.

Method	Relative error	Coverage of 95 % intervals	Wall-clock time [s]
Least squares	2.31	–	0.1
Tikhonov (MAP)	0.12	–	0.2
MCMC (posterior mean)	0.11	0.94	48

3. Computational method. *About 1.5–2.5 pages. Which quantity do you compute (MAP, posterior mean and variance, samples, filtering distribution) and with which algorithm (optimization, Laplace approximation, MCMC, ensemble Kalman, variational inference)? Why this choice for this problem? Give the algorithm in pseudocode if it is not standard, the implementation (language, libraries), the computational cost, and the diagnostics you use (convergence, acceptance rate, effective sample size, trace plots). End with the verification experiment: a synthetic data set with known truth, on which the method must recover what it should. This experiment is mandatory.*

3.1. Algorithm. We explore the posterior (2.8) with ..., summarized in Algorithm 3.1.

3.2. Implementation and diagnostics. The method is implemented in ... using ...; the code is available in the repository linked above. One evaluation of the forward model costs ... Convergence is monitored by ...

3.3. Verification on synthetic data. *Generate data from a known truth with the assumed noise model and show that the method recovers the truth within the stated uncertainty. Report at least one quantitative measure (relative error, coverage of credible intervals) and say what the experiment does and does not verify.*

4. Results. *About 2–3 pages. Describe each experiment so that it can be reproduced: data, parameters, noise level, hyperparameters, random seeds. Show reconstructions with uncertainty (credible bands, posterior samples, marginal densities), compare with a baseline (least squares, Tikhonov, or the method currently used in your field), and show how the results depend on the noise level and the prior hyperparameters. Every figure needs axis labels and a caption that states what is shown and what the reader should notice.*

Figure 1 and Table 1 show ...

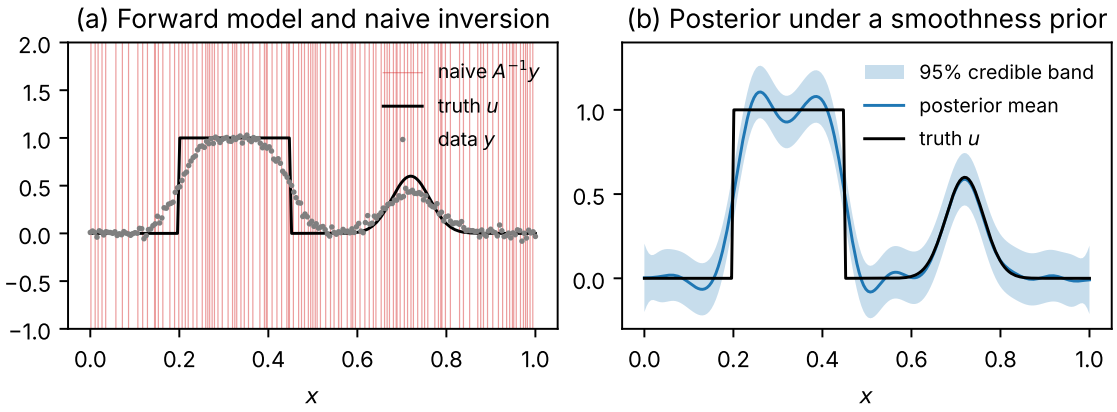


Figure 1: Example figure (one-dimensional deconvolution). (a) Truth, noisy data, and the unstable naive inversion. (b) Posterior mean and 95 % credible band under a smoothness prior. Replace with your own figure.

5. Discussion and outlook. *About 0.5–1 page. What worked and what did not? Which assumptions were most consequential? What are the limitations of the model, the method, and the experiments? What would be needed to turn this into a publication or to use it in your research: more data, a better forward model, a different prior, a faster sampler? Be concrete and honest; a well-argued negative result is a valid outcome.*

Contribution statement. *One or two sentences per author: who did what (modelling, implementation, experiments, writing). Required for teams; omit for solo reports.*

First Author: … Second Author: … Third Author: …

AI-use statement. *Two or three sentences: which generative AI tools you used and for what (literature search, language editing, feedback on drafts, coding assistance), or that none were used. The intellectual content of the report must be yours; you remain responsible for correctness.*

We used … for … All results, derivations, and text were produced and checked by the authors.

Data and code availability. *Link to the reproducibility repository (code, scripts to regenerate every figure, README with installation and execution instructions). State how the data can be obtained; use synthetic or anonymized data if the real data cannot be shared.*

The code and the scripts that reproduce all figures are available at https://github.com/your-team/your-repository. The data …

REFERENCES

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Appendix A. Derivations and additional figures.

Optional: longer derivations, additional experiments, parameter tables. Everything the reader needs to follow the main text belongs in the main text.